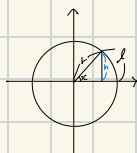




<인공지능 2주차 1차시 수업>

Math
- Tech
- 개념
- 기록록

$$e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n \rightarrow \frac{1}{n-1}, \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^{\frac{1}{n-1}}$$



$$h = r\theta$$

$$S = \frac{1}{2} r \theta = \frac{1}{2} r^2 \theta$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{h}{\frac{1}{2} r \theta} = \lim_{x \rightarrow 0} \frac{h}{\frac{1}{2} x} = 1$$

$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = 1$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$\sin(\theta + \frac{\pi}{2}) = \cos \theta$$

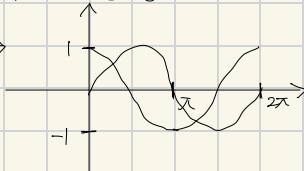
$$\cos(\theta - \frac{\pi}{2}) = \sin \theta$$

$$\sin(\theta + \pi) = -\sin \theta$$



파형은 돌아서만

Real Term만 사용



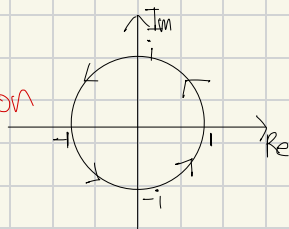
① $\frac{1}{0} \rightarrow 1D$ scalar $\in \text{Real Number}$

② $\frac{(-)}{0} \rightarrow 1D$ Vector $\in \text{Real Number}$

③ $i^2 = -1, i^3 = i^2 \cdot i = -i, i^4 = 1$

↳ 문다

2D Rotation



Real Number

Imaginary Number

$$Z = a + bi: \text{Complex}$$

부정적분

$$\int f(x) dx = F(x) + C$$

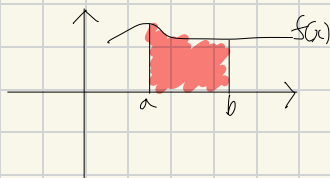
$$\text{ex) } \int \frac{1}{x} dx = \ln x + C, \int \cos x dx = \sin x$$

정적분

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\int \sin x dx = -\cos x, \int \sec^2 x dx = \tan x$$

$$\int \sec(x^2) \tan(x^2) 2x dx = \sec x^2 + C$$



$$y = \ln(\tan(\cos x))$$

$$\int \frac{-\sec^2(\cos x) \cdot \sin x}{\tan(\cos x)} dx = \ln|\tan(\cos x)| + C$$

$$y = \ln(\sec(e^{hx})) \quad y' = \int \frac{\sec(e^{hx}) \cdot \tan(e^{hx}) \cdot e^{hx} \cdot \frac{1}{x}}{\sec(e^{hx})} dx$$

$$\int \frac{5x^4 + 6x + \frac{\cos x}{\sin x}}{x^5 + 3x^2 + \ln(\sin x)} dx = \ln|x^5 + 3x^2 + \ln(\sin x)| + C$$

$$\int f \cdot g dx = f \cdot G - \int f' \cdot G dx \quad \int x \sin dx = -\cos x + \int \cos x dx$$

f. 미분 편한거

g. 적분 "

$$f = x, f' = 1$$

$$g = \sin x, G = -\cos x$$

$$= -x \cos x + \sin x + C$$

$$= -\cos x + (-x)(-\sin x)$$

$$+ \cos x + 0$$

$$= x \sin x$$

100, 200, 300

$$\bar{x} = M_x = E[X_i]$$

$$\text{Avg} = \frac{\sum_{i=1}^N x_i}{N}$$

평차	
100	-100 → 0 ²
200	0
300	100
평균 편차: 0	

음 → 양

① 1

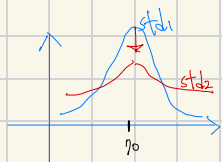
② 0

③ e

$$\sqrt{\text{Var}} = \text{표준편차 (Standard Deviation)} = \text{Std}$$

Variance (분산): 편차의 제곱을 평균

$$\text{Var} = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}$$



Std1 < Std2 → 얼마나만큼 퍼졌나.

$$y = f(t)$$

t=0, 100마리

t=1, 300마리

t=0, ?

$$\frac{df(t)}{dt} = k \cdot f(t)$$

↓

1차 미분방정식

$$\frac{df(t)}{dt} = k \cdot f(t) \Rightarrow \frac{df(t)}{f(t)} = k \cdot dt$$

$$\ln|f(t)| = kt + C$$

$$|f(t)| = e^{kt+C} = e^{kt} \cdot e^C$$

$$f(t) = \pm e^C \cdot e^{kt} = ce^{kt}$$

$$f(t) = ce^{kt}$$

$$\int \frac{1}{f} df = \int k dt \Rightarrow \ln|f| + C_1 = kt + C_2$$

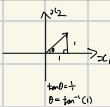
→ scalar → value, metric

$\frac{\partial i}{\partial x} = ?$ $i = C(g) = C\{BCH\} = C\{B[A^{g(x)}]\}$ Loss $\Leftarrow$ Backward Propagation

$$\frac{\partial i}{\partial x} = \frac{\partial i}{\partial g} = \frac{\partial g}{\partial h} = \frac{\partial h}{\partial x}$$

告

$$|x| = \sqrt{2}$$
$$\angle x = \tan^{-1}(1) = \frac{\pi}{4}$$



- ② Vector : $x \in \mathbb{R}^m \Rightarrow x = \begin{bmatrix} x_1 \\ \vdots \\ x_i \\ \vdots \\ x_m \end{bmatrix} \rightarrow x_i \in \mathbb{R}$
($i=1, \dots, m$)

- $$= \begin{bmatrix} x_1 & \dots & x_n \\ \vdots & & \vdots \\ x_m & \dots & x_{nm} \end{bmatrix} \quad x_{ij} = \begin{cases} i=1 \dots m \\ j=1 \dots n \end{cases}$$
- $$= [x_1 \dots x_n] \quad x_n \in \mathbb{R}^m$$
- $$K = n$$

④ Tensor: $x \in \mathbb{R}^{m \times n \times k}$



- $$x \in \mathbb{R}^2 \quad x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \|x\|_2 = 5 \Rightarrow \|x\| = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \quad \text{Norm}$$

- $$x \in \mathbb{R}^n \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$L_1 \text{ norm} = |x| = \sum |x_i| = \|x\|_1$$

$$L_2 \text{ norm} = \|x\|_2 = \sqrt{\sum x_i^2} = \|x\|_2$$

$$\hookrightarrow \|x\|_2^2 = \sum x_i^2 = \|x\|^2$$

$x_1 \cup x_n$

$$W_1x_1 + W_2x_2 + \dots + W_nx_n$$

$$W_1 \cdots W_n = 0$$
 선형 독립

$$W_1 x_1 + W_2 x_2 = 0$$

$$W_1 x_1 = - \underbrace{W_2 x_2}_{W_3}$$

$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\downarrow$

rank $(X) = 2$ (Basis)

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in \mathbb{R}^3$$

full Rankot duple inverse Matrix
ex) if $(k \neq \text{full Rank})$
 $k^{-1} = \text{None}$.

$k^{-1} = \text{None.}$

$$W_1 X_1 + W_2 X_2$$

= Weighted Sum

= Linear Combination (线性组合)

→ 과정을 span이라 함
같은 말

$$x_1 \rightarrow \mathbb{R}^2 = \mathbb{V} \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Span}(x_1, x_2) = \left\{ \frac{w_1 x_1 + w_2 x_2}{\text{Linear C}} \mid w_1, w_2 \in \mathbb{R} \right\}$$

<인공지능 3주차 1차시 수업>

- Col: $x \in \mathbb{R}^m$
- Metric: $x \in \mathbb{R}^{m \times n} \Rightarrow X = \begin{bmatrix} x_{11} & \dots & x_{1n} \\ \vdots & & \vdots \\ x_{m1} & \dots & x_{mn} \end{bmatrix} = [x_1, x_2, \dots, x_n]$
- Tensor: $x \in \mathbb{R}^{m \times n \times k}$

Unit Vector $\rightarrow \mathbb{E}^n \rightarrow \text{Norm}$

$$x \in \mathbb{R}^m \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \quad l_1 = \sum |x_i| = \|x\|_1$$

$$l_2 = \sqrt{\sum x_i^2} = \|x\|_2$$

$$ex) \quad x = \begin{bmatrix} 3 \\ -5 \\ 2 \end{bmatrix}$$

$$l_1 = 10$$

$$l_2 = \sqrt{3^2 + (-5)^2 + 2^2}$$

Supervised Learning

A (정답): label (ground truth)

B (예측)

$$\sum_{i=1}^N |A_i - B_i| = L_1 \text{ loss Function}$$

$$\text{Loss} = \sqrt{\sum_{i=1}^N (A_i - B_i)^2} = L_2$$

parameter $\theta \rightarrow$ 학습을 변경하는 것

alg min Loss Function

$$= \arg \min_{\theta} \left(\sum_{i=1}^N (A_i - B_i) \right)$$

Basis

Span = Linear Combination (선형 조합)

Vector space: 0 (Basis 1개)

Vector space: X

$$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} : x_1, x_2 \text{ Linear Independent}$$

$$\therefore \text{Span}\{x_1, x_2\} = \{w_1 x_1 + w_2 x_2 \mid w_1, w_2 \in \mathbb{R}\} \in \mathbb{R}^2 = \checkmark$$

$\Rightarrow$ Basis Vector 2 Span 2를 채워 Vector space close structure.

$$\begin{bmatrix} 3 \\ 0 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 4 \end{bmatrix} = w_1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + w_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{cases} 2w_1 + w_2 \\ w_2 \end{cases}$$

$$2w_1 + 4 = 3 \quad w_1 = -\frac{1}{2} \quad w_2 = 4$$

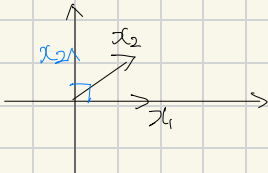
$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Span}(x_1, x_2) \in \mathbb{R}^2 = \checkmark$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Span}(x_1, x_2, x_3) \in \mathbb{R}^3$$

orthogonal



$$w_1 x_1 + w_2 x_2 = w_3 x_3$$

L. indep $\xrightarrow{(0) (x)}$ orthogonal $\xleftarrow{0}$

Shall <Basis Vector는 3개 Linear Indep> = have to = must

$$\mathbb{R}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbb{R}^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$x_1 \quad x_2 \quad x_3$

$$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \text{orthonormal (직각이고 길이 1)}$$

∴ Basis Vector는 span을 이루 Vector space를 생성한다
 → Basis vector는 linear Independent이다.

$$A \in \mathbb{R}^{m \times n}$$

$$AB \Rightarrow C \in \mathbb{R}^{m \times k}$$

$$\begin{pmatrix} m \times n \\ n \times k \end{pmatrix} \Rightarrow \begin{pmatrix} m \times k \end{pmatrix}$$

① Transpose

$$A \in \mathbb{R}^{m \times n}, A^T \in \mathbb{R}^{n \times m}$$

$$\begin{pmatrix} m \times n & n \times m & m=n \\ \left[\begin{matrix} \end{matrix} \right] & = & \left[\begin{matrix} \end{matrix} \right] \\ \text{"rectangular"} & & \text{"square"} \end{pmatrix}$$

$$(AB)^T = B^T \cdot A^T$$

$$(A^T)^T = A$$

$$\sqrt{(A+B)^T = A^T + B^T}$$

② Symmetry (대칭성)

$$A = A^T$$

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

$$A = XX^T$$

$$A = A^T$$

$$(XX^T)^T = (X^T)^T \cdot X^T = XX^T$$

③ Inverse (역행렬)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

full Rank(x)=2 / Linear Independent

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \text{Rank}(A)=3$$

full Rank를 가지는 Inverse Matrix 존재.

⇒ 3 Basis vector들이 linear Independent이다.

$$AB = I \rightarrow \text{정렬행렬 (Identity)} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = A^{-1}$$

$$(AB)^T = B^T \cdot A^T$$

$$(A^{-1})^{-1} = A$$

$$\sqrt{(A+B)^{-1} \neq A^{-1} + B^{-1}}$$

대각행렬

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Zero Matrix

$$x \in \mathbb{R}^m = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}, y \in \mathbb{R}^n = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad n=10, m=5 \quad z \in \mathbb{R}^n$$

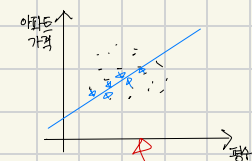
$$z_1 = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + \dots + w_nx_n$$

$$= y = W^T x \in \mathbb{R}^{n \times 1}$$

$$\downarrow$$

$$z = W^T \cdot x = W^T \cdot W^T x$$

<인공지능 3주차 2차시 수업>



x_1, x_2
Linear regression (선형회귀)

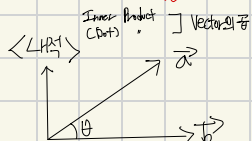
$$Ax=b \iff \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b \end{bmatrix}$$

square

$$Ax=b$$

$$\begin{aligned} A^T \cdot A &= A^T \cdot b \\ \downarrow \text{square} \\ (n \times m)(m \times n) &\end{aligned}$$

$$\therefore x = (A^T \cdot A)^{-1} \cdot A^T \cdot b$$



$$\in \mathbb{R}^2$$

$$\begin{aligned} \vec{a} \cdot \vec{b} &= \langle \vec{a}, \vec{b} \rangle = \|\vec{a}\|_2 \|\vec{b}\|_2 \cos \theta \\ \theta &= \frac{\pi}{2} = \text{orthogonal} \end{aligned}$$

$$\mathbb{R}^{m \times n}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$x \in \mathbb{R}^n, y \in \mathbb{R}^m$$

$$\langle x, y \rangle = x^T \cdot y$$

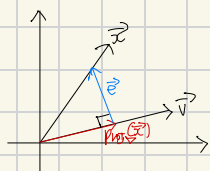
$$= \sum_{i=1}^m x_i y_i$$

$$(x_1, \dots, x_m) \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

$$\langle x, x \rangle = x^T \cdot x = \sum_{i=1}^m x_i^2 = \|x\|_2^2$$

$$\therefore \|x\|_2^2 = x^T x$$

2 projection



$$i) \text{Proj}_{\vec{v}}(\vec{x}) = \alpha \cdot \vec{v} \quad (\alpha \in \mathbb{R})$$

$$ii) \text{Proj}_{\vec{v}} + \vec{e} = \vec{x} \quad \therefore \vec{e} = \vec{x} - \text{Proj}_{\vec{v}}$$

$$iii) \vec{e} \perp \vec{v} \quad \therefore \langle v, e \rangle = \langle e, v \rangle = 0$$

$$\begin{aligned} &= v^T e \\ &= v^T (\vec{x} - \text{Proj}_{\vec{v}}) \\ &= v^T \vec{x} - v^T \text{Proj}_{\vec{v}} \\ &= v^T \vec{x} - \alpha v^T v = 0 \quad \therefore \alpha = \frac{v^T \vec{x}}{v^T v} \end{aligned}$$

$$\text{Proj}_{\vec{v}}(\vec{x}) = \alpha \vec{v}$$

$$\begin{aligned} &= \frac{v^T \vec{x}}{v^T v} \cdot v \\ &= \frac{v v^T}{v^T v} \vec{x} \\ &= \frac{v v^T}{\|v\|_2^2} \vec{x} \end{aligned}$$

Inner Product

3 벡터 편미

$$x = Ax, y \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^n$$

$$\Rightarrow \frac{\partial y}{\partial x} = \frac{\partial Ax}{\partial x} = A$$

$$y = Ax$$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$y_1 = a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n$$

$$y_2 = a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n$$

$$y_m = a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n$$

$$y_i = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n$$

$$(i=1, \dots, m)$$

$$\Rightarrow y_i = f(x_1, x_2, \dots, x_n) \Rightarrow \frac{\partial y_i}{\partial x_1}, \frac{\partial y_i}{\partial x_2}, \dots, \frac{\partial y_i}{\partial x_n}$$

$$\Rightarrow y_m = f(x_1, x_2, \dots, x_n) \Rightarrow \frac{\partial y_m}{\partial x_1}, \frac{\partial y_m}{\partial x_2}, \dots, \frac{\partial y_m}{\partial x_n}$$

$$\begin{bmatrix} a_{i1}, a_{i2}, \dots, a_{in} \\ a_{21}, a_{22}, \dots, a_{2n} \\ \vdots \\ a_{m1}, a_{m2}, \dots, a_{mn} \end{bmatrix} = A$$

Jacobian

$$\textcircled{3} \alpha = x^T A x, A \in \mathbb{R}^{n \times n} \text{ (square)}$$

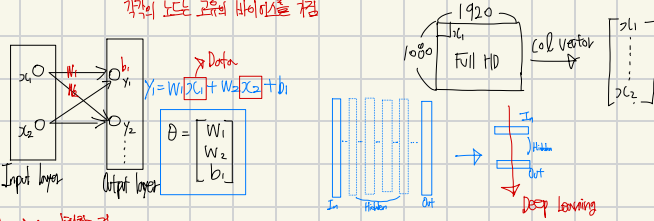
$$\frac{\partial \alpha}{\partial x} = \frac{\partial (x^T A x)}{\partial x} = \frac{\partial (x^T A x)}{\partial x^T} + \frac{\partial (x^T A x)}{\partial x}$$

$$= (Ax)^T + x^T \cdot A$$

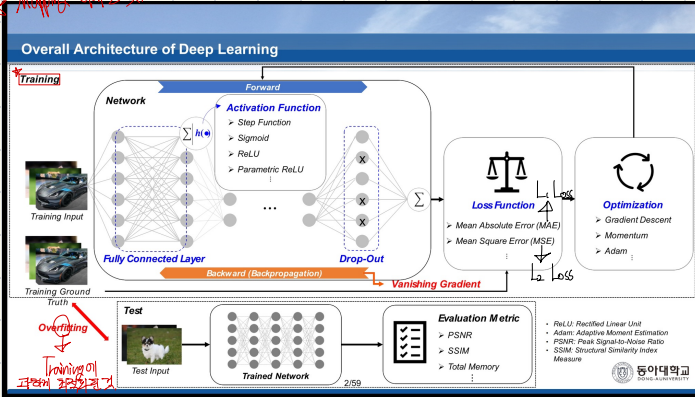
$$= x^T \cdot A^T + x^T \cdot A = x^T (A + A^T)$$

<인공지능 4주차 이론>

Perceptron (인간의 뉴런을 모사) 다중입력 → 단일출력
각각의 노드 코드의 바이너리를 곱함



★ Applying 인공 신경망



Learning

Supervised learning (지도학습)

⇒ 샘플이 있다 ⇒ 답이 있다

↳ Label, Ground Truth

Training Data ↔ Test Data

ex) 95%

85%

90%

90% ✓ Test 업에서 정확도가 높으면 좋음

<문제 해결 방법>

기호의 인코딩 (문자의 인코딩)
컴퓨터가 이해할 수 있는 형태로 변환
ex) Pixel

고양이의 RGB 값과 주변 배경의 RGB 값이 같으면 고양이로 구분 못하는 오류가 생김

↓
XOR 게이트

점 : Inference

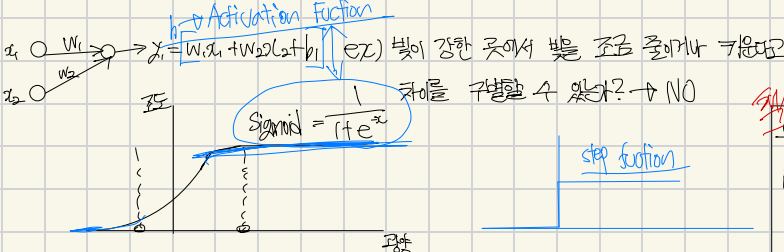
Supervised learning

Regression (Output: Real Number)

Classification

Binary Classification (Output: 0 or 1)

Multi-label (Output: 0, 1, 2, ..., N)



Logic gate



0	0	0
0	1	0
1	0	0
1	1	1



0	0	0
0	1	1
1	0	1
1	1	1



0	1
1	0



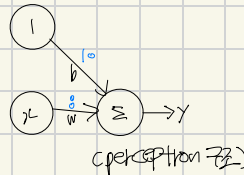
0	0	1
0	1	1
1	0	1
1	1	0



0	0	1
0	1	0
1	0	0
1	1	0



0	0	0
0	1	1
1	0	1
1	1	0



이것이 가지고 있는 것은 어떤 Network가 있을 때
Perceptron을 통해서 w값과 bias값을 구해보자.
어떻게? Loss를 최소화할 수 있는 방법으로
어떤 방법? Backpropagation으로.
Overfitting을 유의하면서.

wt. x
 (conv) (conv)

$$\text{Loss } \sum_{i=1}^n g^2$$

(conv)
 $\mathbb{R}$

$x \in \mathbb{R}^m$ $g \in \mathbb{R}^n$ $b \in \mathbb{R}^n$

$w \in \mathbb{R}^{m \times n}$

$wt \in \mathbb{R}^{n \times n}$

$$\frac{\partial \mathcal{L}}{\partial g} = 2g \in \mathbb{R}^n$$

